

Find Your Integration Partner

TA instructions and optional consultant card

Print pages 1–6 single-sided and cut along the table borders. Each row is a complete pair. For 35 students, remove one full row before shuffling; give the remaining student the consultant card below. For 36 students, use all 18 pairs. For other attendance totals, use complete pairs plus a consultant if needed. Cards repeat deliberately: any correct match is acceptable, and students should form pairs rather than wait for a unique partner.

Give students about two minutes to find a partner and two minutes to introduce themselves, identify du and v , and predict what happens next. They do not need to integrate fully. Keep these partners for the worksheet.

Quick answer key

Type	du	v
$x \cos(4x)$	dx	$\frac{1}{4} \sin(4x)$
$x^3 \ln x$	$\frac{1}{x} dx$	$\frac{x^4}{4}$
$\tan^{-1}(3x)$	$\frac{3}{1+9x^2} dx$	x
$x^2 e^{2x}$	$2x dx$	$\frac{1}{2} e^{2x}$
$e^{2x} \sin x$	$\cos x dx$	$\frac{1}{2} e^{2x}$
$x 3^x$	dx	$\frac{3^x}{\ln 3}$

INTEGRATION CONSULTANT

Join a pair and introduce yourself. Ask your partners to explain why their cards match. Help calculate du and v , and check the signs and constants. Continue the worksheet as a group of three.

INTEGRATION CONSULTANT

Join a pair and introduce yourself. Ask your partners to explain why their cards match. Help calculate du and v , and check the signs and constants. Continue the worksheet as a group of three.

INTEGRATION CONSULTANT

Join a pair and introduce yourself. Ask your partners to explain why their cards match. Help calculate du and v , and check the signs and constants. Continue the worksheet as a group of three.

INTEGRAL

$$\int x \cos(4x) dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = \cos(4x) dx$$

Find someone with the correct match.

INTEGRAL

$$\int x \cos(4x) dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = \cos(4x) dx$$

Find someone with the correct match.

INTEGRAL

$$\int x \cos(4x) dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = \cos(4x) dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^3 \ln x \, dx$$

Find someone with the correct match.

SETUP

$$u = \ln x, \quad dv = x^3 \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^3 \ln x \, dx$$

Find someone with the correct match.

SETUP

$$u = \ln x, \quad dv = x^3 \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^3 \ln x \, dx$$

Find someone with the correct match.

SETUP

$$u = \ln x, \quad dv = x^3 \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int \tan^{-1}(3x) \, dx$$

Find someone with the correct match.

SETUP

$$u = \tan^{-1}(3x), \quad dv = dx$$

Find someone with the correct match.

INTEGRAL

$$\int \tan^{-1}(3x) \, dx$$

Find someone with the correct match.

SETUP

$$u = \tan^{-1}(3x), \quad dv = dx$$

Find someone with the correct match.

INTEGRAL

$$\int \tan^{-1}(3x) \, dx$$

Find someone with the correct match.

SETUP

$$u = \tan^{-1}(3x), \quad dv = dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^2 e^{2x} dx$$

Find someone with the correct match.

SETUP

$$u = x^2, \quad dv = e^{2x} dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^2 e^{2x} dx$$

Find someone with the correct match.

SETUP

$$u = x^2, \quad dv = e^{2x} dx$$

Find someone with the correct match.

INTEGRAL

$$\int x^2 e^{2x} dx$$

Find someone with the correct match.

SETUP

$$u = x^2, \quad dv = e^{2x} dx$$

Find someone with the correct match.

INTEGRAL

$$\int e^{2x} \sin x \, dx$$

Find someone with the correct match.

SETUP

$$u = \sin x, \quad dv = e^{2x} \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int e^{2x} \sin x \, dx$$

Find someone with the correct match.

SETUP

$$u = \sin x, \quad dv = e^{2x} \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int e^{2x} \sin x \, dx$$

Find someone with the correct match.

SETUP

$$u = \sin x, \quad dv = e^{2x} \, dx$$

Find someone with the correct match.

INTEGRAL

$$\int x 3^x dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = 3^x dx$$

Find someone with the correct match.

INTEGRAL

$$\int x 3^x dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = 3^x dx$$

Find someone with the correct match.

INTEGRAL

$$\int x 3^x dx$$

Find someone with the correct match.

SETUP

$$u = x, \quad dv = 3^x dx$$

Find someone with the correct match.